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Thapar Institute of Engineering & Technology, Patiala

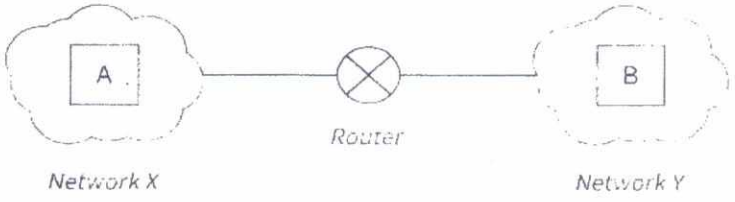
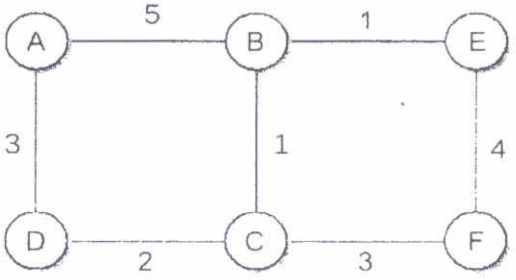
Department of Computer Science and Engineering

AUXILIARY ~~EST~~-EXAM

B. E. (IInd Year): Semester-IV	Course Code: UCS414 /UCS520
Branch: CSE/COE	Course Name: Computer Networks
26th Feb, 2024	Time: 5:30 P.M.
Duration: 3 Hours	Max Marks: 100
Instructor: Dr. Surjit Singh	

Note: *Attempt **ALL** questions in a **proper sequence*** **and keep the subpart of a question at a single place.** *Assume missing data, if any, suitably. Give your answers in detail with proper justification.*

Q1	**i).** An organization X is assigned a class C network address of 201.35.2.0. It uses a netmask of 255.255.255.192 to divide this into sub-networks. Which of the following is/are valid host IP addresses? Writing only answer will not get credit. Please compute and justify. a) 201.35.2.129 b) 201.35.2.191 **ii).** Consider that another organization Y has a class C network address of 201.32.64.0. It uses a subnet mask of 255.255.255.248. Which of the following is NOT a valid broadcast address for any subnetworks? Writing only answer will not get credit. Please compute and justify. a) 201.32.64.135 b) 201.32.64.240 **iii).** Suppose a network with IP Address 192.16.0.0 is divided into four subnets, find the number of hosts per subnet. Also, find the following sub-queries for the second subnet only: Subnet Address, First Host ID, Last Host ID, Broadcast Address. **iv).** In a class B, network on the internet has a subnet mask of 255.255.240.0. Estimate the maximum number of hosts per subnet.	**10**
Q2	**i).** Discuss the behavior of following persistence methods: 1-persistent. Nonpersistent and the *p*-persistent. Also, draw and explain the flow diagrams for the persistence methods. **ii).** In CSMA/CD, identify the relationship among collision detection, propagation delay and the minimum frame size. In a bus CSMA/CD network with a data rate of 100 Mbps, a collision can occur maximum 20 μs after the first bit of the frame leaves the sending station. Compute the minimum length of the frame so that the sender can detect the collision?	**10**
Q3	**i).** Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What can be the maximum bit rate? **ii).** We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need? **iii).** A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity for this channel? **iv).** The SNR is often given in decibels. Assume that SNR(dB) is 36 and the channel bandwidth is 2 MHz. Compute the theoretical channel capacity (MHz).	**10**
Q4	**i).** Give your opinion on QoS by Traffic shaping in network congestion control. Describe the types of traffic shaping. ii). Give the difference between open-loop congestion control and closed-loop congestion control. Name the policies that can prevent congestion and discuss them in detail.	**10**
Q5	**i).** Differentiate between Go-Back ARQ and Stop & Wait Protocols with suitable examples. **ii).** In Selective Repeat protocol, suppose frames through 0 to 4 have been transmitted. Now, imagine that 0 times out, 5 (a new frame) is transmitted, 1 time out, 2 times out and 6 (another new frame) is transmitted. At this point, what will be the outstanding packets in sender's	**10**

	window? Show each step in the scenario at sender and receiver both with the help of neat diagram.	
Q6	**i).** Given the network in below Figure1. In network X, a host named A has an MTU of 520 bytes. In network Y, a host named B has an MTU of 200 bytes. Host A of network X wants to send a message to host B in network Y. Assume a router gets a datagram from host A that contains the length of the header is 20 bytes. How will router look through the datagram and compute number of fragments, value of fragment bits, and offset value?  ***Figure 1*** **ii).** Differentiate between the headers of IPv4 and IPv6.	10
Q7	Consider a network as shown in Figure2, in each round all nodes update their distance vectors using DVR. Compute the initial and final distance vectors of all nodes. Clearly show intermediate steps as well.  ***Figure 2***	10
Q8	**i).** Analyze and discuss the difference between LSR and DVR with the help of above Fig2. **ii).** Show the scheme to represent the data streams 1001010011 using Manchester. Differential Manchester, and 2B1Q. Apply rules mentioned in Table 1 for 2B1Q (Assume that the last signal level has been positive). Table. 1.	10

	Previous level: Positive	Previous level: Negative
Next Bit	Next Level	Next level
00	+3	-3
01	+5	-5
10	-3	+3
11	-5	+5

Q9	Analyze and explain the following in detail: **i).** IEEE 802.11 frame format; **ii).** IEEE 802.3 frame format; **iii).** ARP frame format. **iv).** TCP Header	20